

Hallucination Detection and Evaluation in Multimodal Large Language Models

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Abstract

Multimodal Large Language Models (MLLMs) have demonstrated strong capabilities in visual question answering and image-text reasoning, yet they remain susceptible to hallucinations: outputs that are inconsistent with the visual input, contradict established world knowledge, or contain flaws in the reasoning process. This report presents a systematic framework for detecting and evaluating hallucinations in MLLMs. I adopt a three-category taxonomy covering faithfulness, factuality, and logical hallucinations, and evaluate four representative models (GPT-5.4-mini, Gemini 2.5 Flash, Qwen3.5-35B-A3B, Qwen3-VL-235B-A22B) across three benchmarks (POPE, MathVista, VQA-RAD) using two complementary detection methods: a rule-based pipeline for binary probing and an MLLM-Judge protocol following the MMHal-Bench 0–6 Likert scale, with GPT-5.5 as the primary judge. Three findings emerge. (1) Model rankings reverse across task domains, and a two-axis view of visual-encoder capacity and response-bias calibration explains both the panel ordering on POPE and the ranking shifts on MathVista and VQA-RAD. (2) Chain-of-Thought prompting is baseline-dependent: it substantially reduces hallucinations in weaker-baseline models (GPT-5.4-mini -10.7 pp, Gemini 2.5 Flash -6.7 pp) but exacerbates them in stronger-baseline ones (Qwen3-VL $+7.7$ pp), through two specifically identified mechanisms: an *exposure effect* (latent perceptual errors made visible by reasoning chains) and a *commitment error* (snowballed elaboration around an early wrong premise). (3) In medical VQA, faithfulness hallucinations account for over 96% of all hallucinations, and a recurring confident-but-wrong failure mode (models recite domain conventions accurately yet localize anatomical findings incorrectly) demonstrates that fluent medical knowledge does not entail medical visual perception. Five independent error analyses (human alignment with Cohen’s $\kappa = 0.70$, cross-judge consistency between GPT-5.5 and Claude Opus 4.7 with $\kappa = 0.78$, threshold sensitivity, length-bias assessment, and cross-model overlap) confirm the robustness of the evaluation framework and trace the principal residual judge–human disagreement to a coupling of the informativeness and hallucination dimensions in the underlying scoring rubric.

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1 Introduction

1.1 Task Description

Multimodal Large Language Models (MLLMs) have demonstrated strong capabilities in visual question answering and image-text reasoning, yet they remain susceptible to hallucinations: outputs that are inconsistent with the visual input, contradict established world knowledge, or contain flawed reasoning. This report investigates hallucination detection and evaluation in MLLMs. The assessment requirements are summarized as follows:

1. **Hallucination definition and classification:** define at least two types of hallucinations, provide clear explanations, and supply no fewer than two examples for each type.
2. **Automated detection methods and evaluation metrics:** construct at least two detection methods (e.g., a rule-based method and an MLLM-Judge protocol) and design no fewer than two evaluation metrics.
3. **Experimental evaluation (main task):** evaluate at least two models, including one closed-source and one open-source, on general visual question answering (e.g., POPE) and one of the recommended image-text reasoning benchmarks (MathVista or ChartQA).
4. **Human alignment verification (Human-as-Judge):** construct a small-scale human-annotated set of 20–50 samples as the gold label, compare automated detection results with human judgments, and analyze inconsistent cases.
5. **Analysis requirements:** include an analysis of CoT’s impact on hallucinations, a cross-model comparison, and in-depth analysis of no fewer than two typical failure cases.
6. **Bonus:** extend the evaluation to other image-text tasks or specific domains (e.g., OCR, medical, or remote sensing), or conduct a cross-task generalization analysis.

1.2 Evaluated Models

I evaluate four representative MLLMs that cover both closed-source and open-source models and span a range of architectures (dense and Mixture-of-Experts) and parameter scales (from 3 B to 235 B activated). Two additional models serve as MLLM Judges in the detection pipeline and in the cross-judge consistency check (Appendix A). Table 1 summarizes the key configurations.

Table 1: Model configurations. Four models are evaluated; two additional models serve as MLLM Judges. Architecture lists the total parameter count for MoE models; the active counts are 3 B (Qwen3.5-35B-A3B) and 22 B (Qwen3-VL-235B-A22B).

Model	Role	Architecture	Access Type	API SDK
GPT-5.4-mini (OpenAI, 2026b)	Evaluated	Dense	Closed-source	OpenAI
Gemini 2.5 Flash (Google DeepMind, 2026)	Evaluated	Dense	Closed-source	OpenAI
Qwen3.5-35B-A3B (Qwen Team, 2025a)	Evaluated	MoE 35 B	Open-source	OpenAI
Qwen3-VL-235B-A22B (Qwen Team, 2025b)	Evaluated	MoE 235 B	Open-source	OpenAI
GPT-5.5 (OpenAI, 2026a)	Judge	Dense	Closed-source	OpenAI
Claude Opus 4.7 (Anthropic, 2026)	Judge	Dense	Closed-source	Anthropic

1.3 Evaluation Datasets

I adopt three standard benchmarks that respectively target object-level perception, mathematical visual reasoning, and medical radiology VQA. All datasets are accessed from Hugging Face.¹ Table 2 provides a summary; detailed descriptions are deferred to Appendix A.

Table 2: Dataset overview. POPE samples are subsampled per split for this experiment; MathVista uses the full TESTMINI split and VQA-RAD uses the official test split.

Dataset	Task Type	Samples
POPE (Li et al., 2023)	Object existence probing	3,000 (3 splits $\times$ 1,000)
MathVista (Lu et al., 2024)	Mathematical visual reasoning	1,000 (testmini)
VQA-RAD (Lau et al., 2018)	Medical radiology VQA	451 (test split)

2 Hallucination Definition and Taxonomy

Hallucination remains a central challenge for MLLMs. Although different studies adopt varying terminology and perspectives, a broad consensus has emerged on the underlying concepts. This section reviews three representative surveys (Bai et al., 2025; Chen et al., 2026; Liu et al., 2024a) and then presents the taxonomy adopted in this work.

Bai et al. (2025) define hallucination in MLLMs as generated text that is inconsistent with the corresponding visual content, emphasizing cross-modal inconsistency. They also trace the concept back to LLMs, distinguishing factual hallucinations (deviations from real-world facts) from faithfulness hallucinations (deviations from the input context or self-consistency). Their object-centric taxonomy comprises three levels: category (fabricating or misidentifying objects), attribute (correct category but incorrect properties such as color, shape, or count), and relation (correct objects and attributes but incorrect spatial or interactive relations).

Chen et al. (2026) propose a broader unified definition, treating hallucination as any inconsistency between the generated content and either the input conditions or established world knowledge. Their taxonomy extends the faithfulness-factuality dichotomy: faithfulness hallucinations are stratified by visual granularity into object-level, attribute-level, and scene-level errors, forming a progression from shallow perception to deeper scene understanding; factuality hallucinations are divided into commonsense errors and domain-specific errors.

Liu et al. (2024a) restrict hallucination to the visual domain, defining it as inconsistency between the factual visual content of the image and the generated text. They adopt a dual-perspective taxonomy: a cognitive perspective distinguishing judgment hallucinations (yes/no contradictions with visual facts) from descriptive hallucinations (unfaithful free-form generation), and a semantic perspective distinguishing object, attribute, and relation hallucinations.

Synthesizing these frameworks and given that the experiments in this report primarily target visual question answering and image-text reasoning, I adopt the following definition and three-category taxonomy.

An MLLM hallucination is a model response that is inconsistent with the input visual content, contradicts verifiable real-world knowledge, or contains flaws in the reasoning process.

Table 3 situates this definition relative to the three surveys above. The three categories are detailed below.

¹POPE: <https://huggingface.co/datasets/lmms-lab/POPE>; MathVista: <https://huggingface.co/datasets/AI4Math/MathVista>; VQA-RAD: <https://huggingface.co/datasets/flaviagammarino/vqa-rad>.

Table 3: Comparison of MLLM hallucination taxonomies across three representative surveys and the framework adopted in this work. The bottom row (shaded) marks the taxonomy used throughout this report.

Source	Scope	Primary axis	Sub-categories
Bai et al. (2025)	Cross-modal inconsistency	Factual vs. faithfulness	Category / attribute / relation
Chen et al. (2026)	Generalized inconsistency	Faithfulness vs. factuality	Faithfulness: object / attribute / scene; Factuality: commonsense / domain
Liu et al. (2024a)	Visual domain only	Judgment vs. descriptive	Object / attribute / relation
This work	Visual + knowledge + reasoning	Three orthogonal categories	Faithfulness / factuality / logical

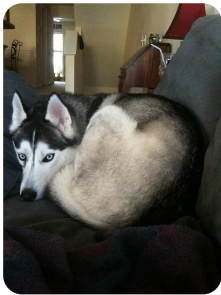
2.1 Faithfulness Hallucination (Visual Inconsistency)

Faithfulness hallucination refers to a model response that is inconsistent with the visual content of the input image. Depending on the level of visual granularity, this category is further divided into three sublevels.

At the object level, the model claims the presence of objects that do not exist in the image, omits objects that are actually present, or misidentifies object categories. The POPE benchmark systematically detects this type of hallucination through binary probing questions (Figure 1a).

At the attribute level, the model correctly identifies the object category but incorrectly describes its properties such as color, shape, material, or pose (Figure 1b).

At the scene level, spatial relations, interactive events, or other scene-level descriptions are inconsistent with the image, even though all individual objects are correctly identified (Figure 1c).



An image of a husky lying on a couch. Object-level faithfulness hallucinations include category substitution (reporting a cat instead of a dog), object omission (claiming no animal is present), and existence misjudgment (affirming the presence of a non-existent object).

(a) Object level.



A round clock mounted on a pole beside a city street. The model correctly identifies the clock but may hallucinate incorrect attributes, such as describing it as square or misreporting the color and hand positions.

(b) Attribute level.



A kitchen countertop with a knife on a cutting board, a tequila bottle to the left of a blender, and a glass to the right. Scene-level hallucinations involve incorrect spatial relations or fabricated interactions between correctly identified objects.

(c) Scene level.

Figure 1: Faithfulness hallucination examples at three levels of visual granularity.

The central detection question for this category is: is the output consistent with the image?

2.2 Factuality Hallucination (Factual Inconsistency)

Factuality hallucination refers to a model response that contradicts established, verifiable real-world knowledge.

At the commonsense level, the error involves everyday knowledge, where the model’s visual perception is roughly correct but the output contradicts well-known facts (Figure 2a).

At the domain-specific level, the error involves specialized professional knowledge. Such hallucinations are particularly common in the VQA-RAD dataset (Figure 2b).



An image showing zebras and giraffes. Commonsense factuality hallucinations arise when the model’s roughly correct visual perception leads to outputs that contradict basic biological knowledge, such as misidentifying a zebra as a painted horse.

(a) Commonsense level.



A chest X-ray with no obvious abnormalities in the lung fields. Domain-specific factuality hallucinations occur when the model fabricates findings such as pleural effusion or cardiomegaly that contradict the actual radiological evidence.

(b) Domain-specific level.

Figure 2: Factuality hallucination examples at the commonsense and domain-specific levels.

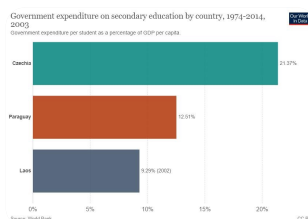
The central detection question for this category is: is the output consistent with world knowledge?

2.3 Logical Hallucination (Reasoning Error)

Logical hallucination refers to a model response with flaws in the reasoning process, where the conclusion does not logically follow from the visual evidence or from its own established premises.

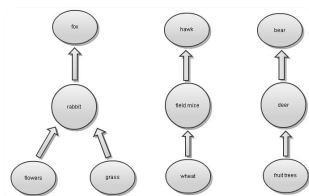
In a deductive reasoning error, the model correctly reads the visual information but reaches a contradictory conclusion through faulty inference. Such errors, particularly numerical reasoning failures, are the primary source of logical hallucinations in MathVista (Figure 3a).

In a causal reasoning error, the model infers causal relationships that contradict the visual evidence (Figure 3b).



A bar chart showing government expenditure percentages for Czechia, Paraguay, and Laos. The model correctly reads the bar values but errs in the comparison step, producing a conclusion that contradicts its own extracted evidence.

(a) Deductive reasoning.



A food web showing predation relationships among field mice, snakes, hawks, grasshoppers, frogs, and grass. Causal reasoning hallucinations arise when the model incorrectly traces causal chains, reversing the expected direction of population effects.

(b) Causal reasoning.

Figure 3: Logical hallucination examples in deductive and causal reasoning.

The central detection question for this category is: is the reasoning valid?

3 Experimental Design

3.1 Detection Methods and Evaluation Metrics

3.1.1 POPE: Object-Existence Probing

The detection method for POPE is rule-based matching.

Prompt and post-processing. During response generation, the model receives a prompt containing only the question and an instruction to answer Yes or No, ensuring concise and parseable responses (the complete prompt template is provided in Appendix A). The answer post-processing algorithm follows the official evaluation script of the RUCAIBox/POPE repository;² the pseudocode is given in Algorithm 1.

Algorithm 1 POPE answer normalization

Require: Model response r , a free-form text string

Ensure: Normalized answer $a \in \{\text{“yes”}, \text{“no”}\}$

```
1:  $s \leftarrow$  first sentence of  $r$  // split by “.”
2: remove all commas from  $s$ 
3:  $W \leftarrow$  split  $s$  by whitespace into a set of words
4: if “No”  $\in W$  or “not”  $\in W$  or “no”  $\in W$  then
5:   return “no”
6: end if
7: return “yes”
```

Object hallucination in POPE. Under this protocol, the model commits an object hallucination whenever it answers “yes” while the ground-truth label is “no.” The event is a false positive in the binary-classification view and instantiates the object-level faithfulness category of the taxonomy.

The metrics treat “yes” (object present) as the positive class, so object hallucinations correspond to false positives. The core metrics are summarized in Table 4.

Table 4: POPE evaluation metrics. Hallucination Rate is the core hallucination metric.

Metric	Formula	Description
Accuracy	$(TP + TN)/N$	Overall correctness
Precision	$TP/(TP + FP)$	Accuracy when predicting Yes
Recall	$TP/(TP + FN)$	Detection rate for actual Yes
F1	$2PR/(P + R)$	Harmonic mean of Precision and Recall
Yes Ratio	N_{yes}/N	Response bias indicator
Hallucination Rate	FP/N	Core metric; equals object hallucination rate

3.1.2 MathVista: Mathematical Visual Reasoning

Detection method. The detection method for MathVista employs an MLLM Judge following the MMHal-Bench evaluation protocol (Sun et al., 2024), which adopts a 0–6 Likert scale that jointly assesses the informativeness and the hallucination level of each response. A threshold of 3 separates the two regimes: scores below 3 indicate hallucination, and scores at or above 3 indicate no hallucination. GPT-5.5 serves as the Judge in this implementation. The judge’s system prompt defines the scoring rubric, the hallucination criteria, and a three-way type classification, and is supplemented by five few-shot examples for consistency. Whenever the judge assigns a score below 3, it additionally labels the hallucination type (faithfulness, factuality, or

²<https://github.com/RUCAIBox/POPE/blob/main/evaluate.py>

logical). The complete prompt templates for both response generation and judge scoring are provided in Appendix A.

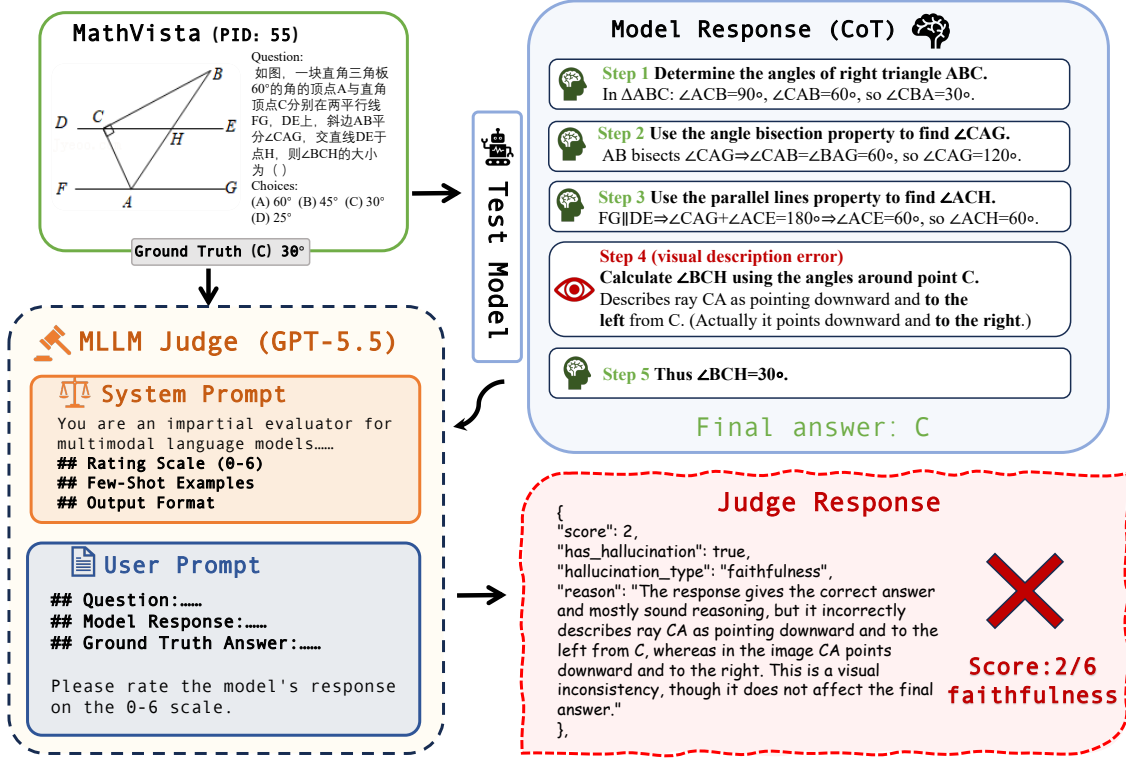


Figure 4: MLLM Judge evaluation framework for MathVista: the judge receives a question, a model response, and a scoring rubric, and produces a score (0–6) with an optional hallucination type label.

Prompt construction. During response generation, prompts are dynamically constructed from sample metadata to produce format-specific instructions. In Direct mode, the prompt consists of the question text concatenated with format instructions. In CoT mode, the prompt further requires step-by-step reasoning with the final answer marked by “Final answer:”. Both modes are evaluated to analyze the effect of explicit reasoning on hallucination behavior.

The evaluation metrics are shown in Table 5.

Table 5: MathVista evaluation metrics. Hallucination Rate is the core hallucination metric.

Metric	Definition	Description
Hallucination Rate	$N_{\text{score}<3}/N_{\text{total}}$	Lower is better
Average Score	$\sum \text{score}_i/N$	Overall response quality
Type-specific count	By faithfulness/factuality/logical	Hallucination source breakdown

3.1.3 VQA-RAD: Medical Radiology VQA

Medical adaptation. The detection method for VQA-RAD reuses the MLLM Judge protocol (0–6 Likert scale, threshold 3) with domain-specific prompt adaptations for the medical setting (see Appendix A). The judge’s task description is replaced with a radiology-specific scenario description, requiring the judge to verify consistency with both the image’s visual content and established medical and anatomical knowledge. During response generation, a radiologist expert role prompt is adopted following the role-setting strategy of Wu et al. (2024). The

three-way hallucination type classification (faithfulness, factuality, logical) is always enabled, and hallucination rates are additionally stratified by closed-ended and open-ended answer types.

The full 451-sample test split is used, comprising 251 closed-ended (yes/no) and 200 open-ended (free-text) questions. The evaluation metrics include Hallucination Rate, Average Score, type-specific hallucination counts, and hallucination rates stratified by answer type.

3.2 Experimental Conditions

All four models are accessed through APIs with a unified random seed of 42. Temperature is kept at each platform’s default, and the maximum token length is uncapped. POPE samples 1,000 items per split, MathVista uses the full testmini set (1,000 items), and VQA-RAD uses the test split (451 items). GPT-5.5 serves as the MLLM Judge throughout.

Table 6 summarizes the correspondence among detection methods, datasets, and metrics.

Table 6: Overview of method–dataset–metric correspondence. The shaded bottom row reports the human alignment study conducted on a 40-sample MathVista CoT subset (Appendix A).

Dataset	Detection Method	Annotation Scope	Core Metrics
POPE	Rule-based	Faithfulness	HR, F1
MathVista	MLLM Judge	All three types	HR, Avg Score
VQA-RAD	MLLM Judge	All three types	HR, Avg Score, Closed/Open HR
Human	Manual annotation	All three types	Kappa, F1, Type Agreement

4 Experiment 1: POPE Object Hallucination Detection

4.1 Results

Table 7 reports all evaluation metrics for the four models across the three POPE splits. Columns are grouped by split, each containing Accuracy, Precision, Recall, F1, Yes Ratio, Hallucination Rate (HR), and the four confusion matrix entries (TP/TN/FP/FN).

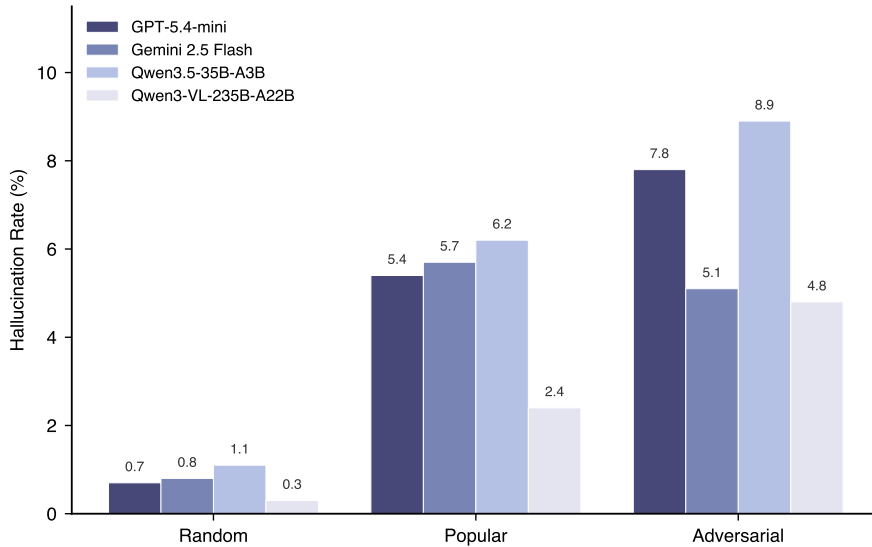


Figure 5: Hallucination rates of four MLLMs across the three POPE splits (Random, Popular, Adversarial).

Table 7: Comprehensive POPE results across three splits. Best values per metric are highlighted in pink.

Split	Metric	GPT-5.4-mini	Gemini 2.5 Flash	Qwen3.5-35B	Qwen3-VL-235B
Random	Accuracy	0.9010	0.9160	0.9260	0.9200
	Precision	0.9831	0.9814	0.9754	0.9929
	Recall	0.8156	0.8477	0.8737	0.8457
	F1	0.8916	0.9097	0.9218	0.9134
	Yes Ratio	0.414	0.431	0.447	0.425
	HR	0.70%	0.80%	1.10%	0.30%
	TP/TN/FP/FN	407/494/7/92	423/493/8/76	436/490/11/63	422/498/3/77
Popular	Accuracy	0.8570	0.8620	0.8730	0.8970
	Precision	0.8836	0.8800	0.8750	0.9459
	Recall	0.8216	0.8377	0.8697	0.8417
	F1	0.8515	0.8583	0.8724	0.8908
	Yes Ratio	0.464	0.475	0.496	0.444
	HR	5.40%	5.70%	6.20%	2.40%
	TP/TN/FP/FN	410/447/54/89	418/444/57/81	434/439/62/65	420/477/24/79
Adversarial	Accuracy	0.8280	0.8730	0.8520	0.8750
	Precision	0.8385	0.8924	0.8318	0.8979
	Recall	0.8116	0.8477	0.8818	0.8457
	F1	0.8248	0.8695	0.8560	0.8710
	Yes Ratio	0.483	0.474	0.529	0.470
	HR	7.80%	5.10%	8.90%	4.80%
	TP/TN/FP/FN	405/423/78/94	423/450/51/76	440/412/89/59	422/453/48/77

4.2 Analysis

4.2.1 Three Splits Probe Three Error Mechanisms

Monotonic difficulty gradient. From Random to Popular to Adversarial, the Hallucination Rate of all four models increases monotonically (Table 7 and figure 5), confirming the validity of POPE’s three-tier negative-sampling design. Beyond a difficulty ranking, however, the three splits also expose three qualitatively distinct error mechanisms that map cleanly onto the faithfulness taxonomy.

Random split: object-level perception. Negative objects are sampled uniformly, so most candidates are visually unrelated to the scene (e.g., asking about a car in a kitchen photo). All four models achieve HR between 0.30% and 1.10%, indicating that the baseline ability to reject obviously absent objects is near saturation. This split therefore measures raw object-level visual perception, with negligible interference from language priors.

Popular split: resistance to linguistic priors. Negative objects are high-frequency co-occurring categories that are statistically associated with the scene but absent from the specific image (e.g., asking about a backpack in an outdoor eating scene). HR rises sharply to 2.40%–6.20%, roughly 5–8 times the Random level. The surge isolates the contribution of language-prior bias: when the visual evidence is ambiguous, models fall back on the statistical co-occurrence patterns learned during pretraining. Case 2 in Section 4.3.2 provides a direct instance of this mechanism.

Adversarial split: fine-grained semantic discrimination. Negative objects are semantically similar to objects that do appear in the image (e.g., asking about a bear when only a teddy bear is present). HR rises further to 4.80%–8.90% and the model can no longer rely on scene-level associations: the visual encoder must resolve fine-grained semantic boundaries. Case 1 in Section 4.3.1 shows that this is the hardest of the three failure modes, because the underlying cause, confusion of conceptual hierarchies, is not addressable by scale alone.

In sum, the three splits jointly diagnose where in the perception–prior pipeline an object-level faithfulness hallucination originates, rather than merely how often it occurs.

4.2.2 Cross-Model Comparison

Within the Qwen family, visual-encoder capacity dominates. Qwen3-VL-235B achieves the lowest HR on every split (0.30%, 2.40%, 4.80%) and the highest F1 on Popular and Adversarial. Compared with Qwen3.5-35B-A3B, which shares the same training family but uses a smaller activated visual encoder (3 B vs. 22 B activated parameters), HR is consistently 50–75% lower despite identical instruction-tuning regimes. Holding the language backbone family fixed, visual-encoder scale is therefore the dominant factor within this family; whether the same holds across families cannot be established here, because the closed-source dense models do not disclose visual-encoder parameters.

GPT-5.4-mini: conservative strategy with asymmetric costs. GPT-5.4-mini has the lowest Yes Ratio of all models (0.414 on Random) and the highest Precision (0.9831 on Random). This conservative profile yields a low FP count but the highest FN count (92 on Random), so its hallucinations manifest predominantly as omissions rather than fabrications. In safety-sensitive applications this asymmetric cost may be desirable (it is generally preferable not to invent objects), but it caps the achievable F1: HR on Adversarial rises more than tenfold relative to Random (0.70% $\rightarrow$ 7.80%) once the Yes/No decision can no longer be safely defaulted to “No.”

Gemini 2.5 Flash: atypical Popular–Adversarial inversion. Gemini is the only model whose HR *decreases* from Popular (5.70%) to Adversarial (5.10%), though the gap is small (0.6 pp). Since the standard interpretation (Adversarial is harder than Popular) holds for the other three models, this inversion suggests that Gemini’s visual perception is comparatively robust to fine-grained semantic distractors but relatively sensitive to scene-level co-occurrence priors. A plausible mechanism is a perception pipeline that grounds responses more tightly in visible features than in scene context, which would also be consistent with its moderate but stable Popular performance.

Qwen3.5-35B: largest split-difficulty slope. Qwen3.5-35B-A3B exhibits the steepest HR increase across splits (1.10% $\rightarrow$ 6.20% $\rightarrow$ 8.90%, a 7.8 pp rise), the largest in the panel. With only 3 B activated visual parameters, the visual encoder is insufficient to support fine-grained discrimination once the negative samples become semantically adjacent. The high Recall (highest in three of three splits) reflects an opposite pressure toward answering “Yes,” which is also visible in the panel’s highest Yes Ratio (0.529 on Adversarial).

Synthesis. Whereas the previous subsection diagnosed where in the perception–prior pipeline an error originates, the cross-model view identifies who is vulnerable to each mechanism. The comparison highlights two axes: visual-encoder capacity, which explains the gap between the two Qwen models and the rough position of Gemini; and response-bias calibration, which separates the conservative profile of GPT-5.4-mini from the more permissive profiles of the Qwen models. Models near the corner of high capacity and balanced calibration (Qwen3-VL) dominate the leaderboard; modest capacity with permissive calibration (Qwen3.5) sacrifices precision under adversarial pressure; and modest capacity with conservative calibration (GPT-5.4-mini) sacrifices recall. This two-axis view will recur in Sections 5 and 6 under different task pressures.

4.3 Failure Case Analysis

4.3.1 Case 1: Semantic Granularity Confusion (Adversarial)

This case instantiates the fine-grained semantic discrimination mechanism identified in Section 4.2.1. Table 8 and figure 6 present a sample from the Adversarial split. The image contains

a teddy bear, a person, and a book. The question asks whether there is a bear in the image, and the ground truth is No. All four models answer Yes, indicating a systematic error rather than an individual model deficiency.

Table 8: POPE Case 1: sample information and model responses.

Sample Information	
Image	COCO_val2014_000000567886
Objects present	teddy bear, person, book
Objects absent	bear, car, tv
Question	Is there a bear in the image?
Ground Truth	No
Model Responses	
GPT-5.4-mini	Yes
Gemini 2.5 Flash	Yes
Qwen3.5-35B-A3B	Yes
Qwen3-VL-235B	Yes



Figure 6: POPE Case 1: a teddy bear on a couch.

The root cause is semantic granularity confusion. The image contains a teddy bear, an independent category in the COCO label set (Lin et al., 2014), while the question asks about a bear (a real animal). All four models associate “teddy bear” with “bear” at the semantic level, triggering a positive judgment upon seeing the toy. This is essentially conceptual hierarchy confusion: models conflate visually similar objects with semantically identical ones. This error cannot be resolved by increasing model scale or improving visual encoder resolution, as it concerns the clarity of conceptual boundaries in semantic representations rather than the discriminability of low-level visual features.

4.3.2 Case 2: Linguistic Prior Bias (Popular)

This case exemplifies the linguistic-prior bias mechanism isolated by the Popular split (Section 4.2.1). Table 9 and figure 7 present a sample from the Popular split. The image shows a person eating a sandwich and a hot dog outdoors. The question asks whether there is a backpack in the image, and the ground truth is No. Again, all four models answer Yes.

Table 9: POPE Case 2: sample information and model responses.

Sample Information	
Image	COCO_val2014_000000569839
Objects present	person, sandwich, hot dog
Objects absent	car, bottle, backpack
Question	Is there a backpack in the image?
Ground Truth	No
Model Responses	
GPT-5.4-mini	Yes
Gemini 2.5 Flash	Yes
Qwen3.5-35B-A3B	Yes
Qwen3-VL-235B	Yes



Figure 7: POPE Case 2: an outdoor dining scene.

The root cause is linguistic prior bias. In outdoor scenes with people eating, “backpack” is a high-frequency co-occurring object. The statistical prior in the language module (outdoor + person + eating $\rightarrow$ high backpack probability) overrides the visual module’s actual perception, causing the model to produce an affirmative answer. This is precisely the mechanism that the Popular split exploits: models fill visual uncertainty with linguistic priors from training data. The contrast between the two cases further confirms that the Popular and Adversarial splits target distinct error mechanisms, with Popular primarily exploiting linguistic priors and Adversarial primarily testing semantic discrimination.

5 Experiment 2: MathVista Mathematical Reasoning Hallucination Detection

5.1 Direct Mode: Results

Table 10 reports the MathVista evaluation results for all four models in Direct mode.

Table 10: MathVista Direct mode results. Best values are highlighted in pink.

Metric	GPT-5.4-mini	Gemini 2.5 Flash	Qwen3.5-35B	Qwen3-VL-235B
Hallucination Rate	45.1%	34.3%	13.3%	16.3%
Average Score	3.31	4.46	5.00	4.85
$n_{\text{hallucinated}}$	451	343	133	163
Faithfulness	189	278	98	113
Factuality	14	13	5	6
Logical	248	52	30	44
None	549	657	867	837

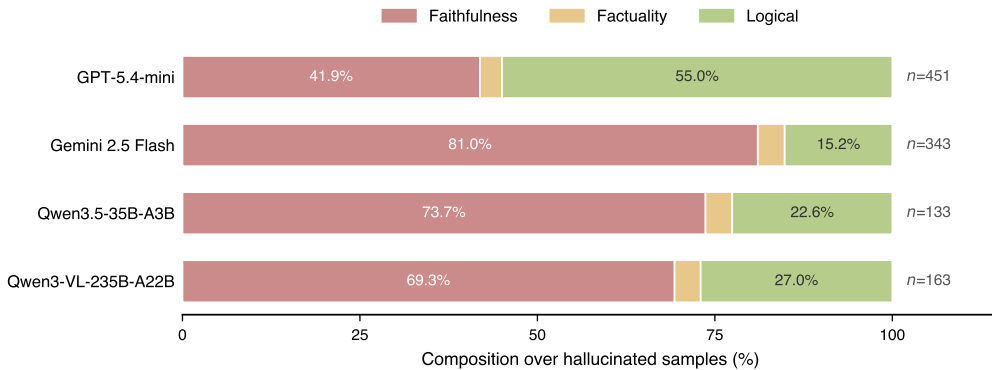


Figure 8: Hallucination type composition over hallucinated samples only in MathVista Direct mode. Each bar sums to the model’s total hallucinated count n .

5.2 Direct Mode: Analysis

From POPE to MathVista: a richer error surface. On POPE the four-model HR range spans roughly 5 percentage points; on MathVista it spans more than 30 points (13.3%–45.1%), because the task simultaneously probes visual perception, mathematical knowledge, and multi-step reasoning. From the worst performer GPT-5.4-mini (45.1%) to the best Qwen3.5-35B-A3B (13.3%), HR differs by a factor of 3.4, far exceeding the cross-model gap on POPE. The capacity–calibration view from Section 4.2.2 remains useful as a starting point, but each model also exposes a distinct dominant error type that cannot be read off POPE alone.

Distinct dominant error types. GPT-5.4-mini is dominated by logical hallucinations (55.0%, 248 out of 451). In Direct mode this model typically outputs only an option letter (e.g., “B”) with a median response length of just one word, providing no reasoning process for the judge to assess. Many incorrect answers are thus directly classified as logical hallucinations, reflecting both weak mathematical reasoning and an ultra-short response style that offers no reasoning chain to earn partial credit.

Gemini 2.5 Flash is dominated by faithfulness hallucinations (81.0%, 278 out of 343). This model produces highly detailed responses (median 242 words) with extensive visual descriptions, but the descriptions contain visual misreadings, such as incorrectly reporting chart values, colors, or labels. Gemini’s weakness lies not in reasoning (only 52 logical cases) but in imprecise visual perception: describing too much leads to getting too much wrong.

Both Qwen models achieve hallucination rates less than one-third of GPT-5.4-mini. Qwen3.5-35B-A3B (13.3%) and Qwen3-VL-235B (16.3%) also show a more balanced distribution across the three hallucination types, with no obvious single weakness. This contrasts with POPE results, where the smaller Qwen3.5 (3 B activated) trails the 235 B model slightly, but on MathVista’s mathematical reasoning it leads by a small margin, suggesting higher efficiency in mathematical reasoning despite its smaller activated parameter count.

5.3 CoT Mode: Results and Comparison

Table 12 compares the core metrics between Direct and CoT modes, including overall performance changes and per-type hallucination breakdowns.

Table 11: MathVista Direct vs. CoT hallucination rates with 95% confidence intervals and paired significance tests. All models share the same $N = 1000$ MathVista testmini items across both conditions, so a paired McNemar test is appropriate.

Model	Direct HR [95% CI]	CoT HR [95% CI]	Δ HR [95% CI]	McNemar p
GPT-5.4-mini	45.1% [42.0, 48.2]	34.4% [31.5, 37.4]	−10.7 pp [−14.1, −7.3]	1.9×10^{-9}
Gemini 2.5 Flash	34.3% [31.4, 37.3]	27.6% [24.9, 30.5]	−6.7 pp [−9.5, −3.9]	6.1×10^{-6}
Qwen3.5-35B	13.3% [11.3, 15.5]	15.5% [13.4, 17.9]	+2.2 pp [+0.5, +3.9]	0.016
Qwen3-VL-235B	16.3% [14.1, 18.7]	24.0% [21.5, 26.7]	+7.7 pp [+5.3, +10.1]	1.0×10^{-9}

Note: Wilson 95% CIs for HR; Wald 95% CI for the paired difference Δ HR. McNemar p -values use continuity-corrected χ^2 (df=1).

Table 12: MathVista Direct vs. CoT: average score and per-type hallucination count changes.

Model	Direct Avg	CoT Avg	Δ Avg	Δ Faith.	Δ Fact.	Δ Logical	Net improvement
GPT-5.4-mini	3.31	4.35	+1.04	+8	−10	−105	+107
Gemini 2.5 Flash	4.46	4.81	+0.35	−66	−8	+7	+67
Qwen3.5-35B	5.00	5.34	+0.34	+26	−3	−1	−22
Qwen3-VL-235B	4.85	4.94	+0.09	+41	0	+36	−77

Note: Per-type columns report the change in raw count (CoT minus Direct) for each hallucination category; negative values indicate fewer hallucinations under CoT. Net improvement is the signed change in total hallucinated samples (positive = improvement).

5.4 CoT Effect Analysis

5.4.1 Baseline-Dependent CoT Effects

CoT effects on MathVista divide cleanly along a single axis: the Direct-mode baseline. Models with high Direct HR (GPT-5.4-mini at 45.1%, Gemini at 34.3%) improve substantially under

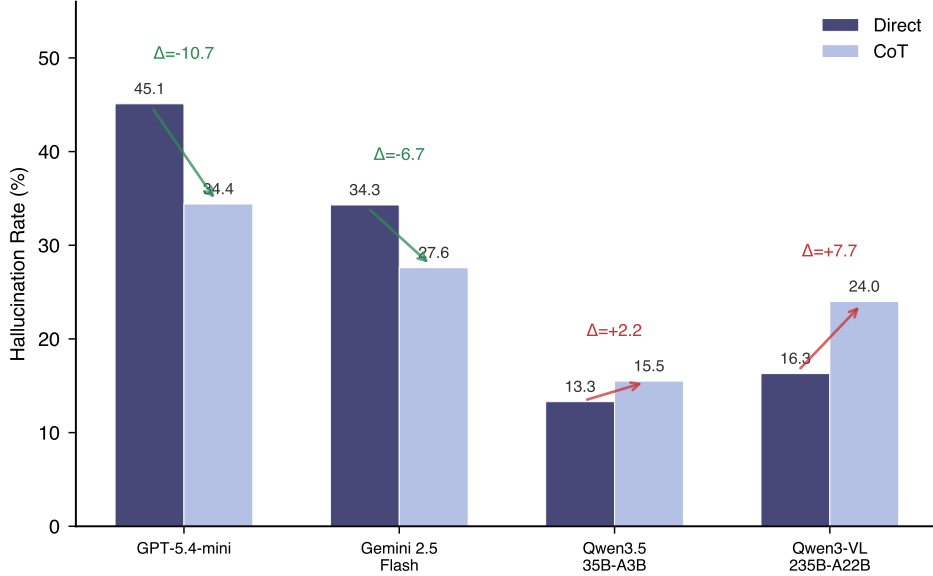


Figure 9: Comparison of hallucination rates between Direct and CoT modes. Δ values indicate the change.

CoT, whereas models with low Direct HR (Qwen3.5 at 13.3%, Qwen3-VL at 16.3%) degrade. All four shifts are statistically significant under a paired McNemar test (Table 11), though the p-values differ by several orders of magnitude: the GPT and Qwen3-VL effects reach $p < 10^{-9}$, while the Qwen3.5 +2.2pp shift is far weaker ($p = 0.016$, 95% CI [+0.5, +3.9]). Table 12 and figure 9 quantify the accompanying score and per-type changes. The remainder of this subsection isolates the mechanisms responsible for each direction and identifies why CoT helps logical hallucinations but exacerbates faithfulness hallucinations.

5.4.2 CoT Beneficiaries: Weaker Baselines

CoT significantly improves weaker-baseline models. GPT-5.4-mini is the largest beneficiary: HR drops by 10.7 percentage points and average score increases by 1.04, both the largest improvements among all models. The improvement comes primarily from a sharp reduction in logical hallucinations ($248 \rightarrow 143$, -42.3%), as shown in Table 12. In Direct mode this model outputs only option letters (median one word), exposing incorrect answers directly to the judge without any reasoning evidence. CoT forces the model to organize its mathematical derivation step by step, correcting many errors that a reasoning chain can catch. Sample-level analysis shows that 209 Direct-mode hallucinations are corrected under CoT, while only 102 cases exhibit reverse degradation, yielding a net improvement of 107 samples.

Gemini 2.5 Flash improves through a different pathway: the benefit comes primarily from reduced faithfulness hallucinations ($278 \rightarrow 212$, -23.7%). CoT requires the model to explicitly cite visual evidence during reasoning, forcing more careful examination of chart content and reducing visual misreadings. Notably, Gemini’s Direct-mode responses are already highly detailed (median 242 words), and its CoT responses are actually slightly shorter (median 228 words), indicating that CoT does not force longer output but rather improves accuracy by constraining reasoning to image-visible evidence.

5.4.3 CoT Risks: Stronger Baselines

CoT introduces additional risk for stronger-baseline models. Both Qwen models exhibit a polarization pattern: 6-score and 2-score bins both grow while intermediate 4–5 scores collapse,

indicating that CoT pushes responses toward the extremes of either perfect correctness or detailed incorrectness.

Qwen3.5-35B-A3B exhibits a pattern in which HR rises by 2.2 percentage points while average score also rises from 5.00 to 5.34. Score polarization explains both moves: 6-score samples grow from 488 to 810 (lifting the mean), and 2-score samples grow from 125 to 154 (detailed but incorrect reasoning is strictly penalized, lifting HR). Since this model already possesses strong mathematical reasoning in Direct mode (only 30 logical cases), CoT yields little marginal benefit and instead exposes blind spots in visual perception by forcing explicit visual description, causing faithfulness hallucinations to rise from 98 to 124 (+26 cases, +26.5%).

Qwen3-VL-235B shows a more pronounced pattern: HR jumps from 16.3% to 24.0% (+7.7pp), with a net degradation of 77 samples. Of these, 116 Direct-mode correct samples produce hallucinations under CoT, while only 39 improve in the reverse direction. The polarization effect is equally pronounced: 6-score samples surge from 452 to 723, but 2-score samples increase from 142 to 194 and 0-score cases appear ($0 \rightarrow 8$), while intermediate 4–5 scores nearly vanish ($93 \rightarrow 6$ and $291 \rightarrow 30$ respectively). Faithfulness hallucinations increase from 113 to 154 (+41 cases, +36.3%), indicating that forcing explicit visual description systematically exposes perceptual weaknesses.

5.4.4 Underlying Mechanisms: Exposure and Commitment

This phenomenon aligns with recent findings. Liu et al. (2025) show through attention analysis that as reasoning chains grow longer, the model’s attention to visual input weakens (attention drift), gradually shifting from image content to linguistic priors, leading to “more thinking, less seeing.” Kancheti et al. (2025) report, after evaluating 17 models, that CoT consistently degrades visual spatial reasoning, with models “hallucinating visual details from textual priors.” The substantial increase in faithfulness hallucinations for both Qwen models is consistent with both observations.

CoT introduces hallucinations through two complementary mechanisms.

Exposure effect. Visual perception errors that remain hidden in Direct mode are explicitly described in the CoT reasoning chain and captured by the MLLM Judge. Even when the final answer is correct, visual misreadings in intermediate steps lower the score. This mechanism explains why stronger-baseline models see increased HR despite high final-answer accuracy.

Commitment error. CoT forces the model to make explicit visual or mathematical assertions early in the reasoning process. Once an intermediate step goes wrong, the model continues reasoning around the incorrect premise, producing “informative but wrong” responses (score 2). The Multimodal Hallucination Snowballing framework proposed by Zhong et al. (2024) directly corresponds to this mechanism: once the model produces a hallucination early in reasoning, it elaborates coherently around the erroneous premise, causing the error to propagate and amplify along the chain.

These two mechanisms operate together: the exposure effect reveals latent perceptual errors, while the commitment error transforms them into cascading reasoning failures. Together they explain why CoT can simultaneously improve weaker models (by catching logical errors through structured reasoning) and degrade stronger models (by forcing explicit articulation of visual misperceptions that would otherwise remain implicit and harmless).

5.5 Failure Case Analysis

This section presents two typical failure cases, each corresponding to one of the two core CoT side-effect mechanisms.

5.5.1 Case 1: CoT Commitment Error Leading to Logical Hallucination (GPT-5.4-mini, PID=30)

Table 13 presents the sample information, and Figure 10 shows the accompanying image. The question is a circle geometry problem: two chords AB and CD intersect at E , with $\angle D = 35^\circ$ and $\angle AEC = 105^\circ$; the task is to find $\angle C$. The correct answer is 70° (option B).

Table 13: MathVista Case 1: sample information.

Field	Content
Question type	Geometry reasoning (multiple choice)
Question	Two chords AB , CD intersect at E ; $\angle D = 35^\circ$. Find $\angle C$.
Ground Truth	70° (B)

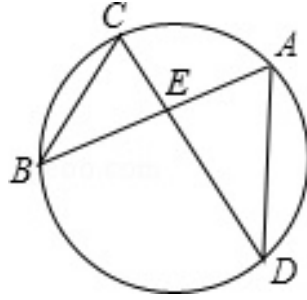


Figure 10: MathVista Case 1: intersecting chords in a circle.

Table 14: MathVista Case 1: responses under two modes.

Mode	Response	Score	Type
Direct	B (70°)	5	none
CoT	Flawed reasoning, arrives at 80° (C)	1	logical

In Direct mode the model outputs the correct option B (70°) without showing any reasoning, and the judge assigns score 5. In CoT mode the model attempts step-by-step reasoning but commits to an incorrect premise in the first step, treating $\angle E = 105^\circ$ as a given condition and applying the triangle angle-sum formula to obtain $\angle C = 180^\circ - 105^\circ - 35^\circ = 40^\circ$. Finding that 40° is not among the choices, the model then fabricates a supplementary-angle argument to produce 80° and selects option C. This exhibits the snowballing hallucination pattern described by [Zhong et al. \(2024\)](#): once an incorrect premise is locked in, subsequent steps build upon it, and each step appears plausible in isolation while the entire chain rests on a false initial assumption.

The correct approach uses the inscribed angle theorem: $\angle AEC$ is the vertical angle of $\angle DEB$, and $\angle C = \angle AEC - \angle D = 105^\circ - 35^\circ = 70^\circ$. In Direct mode the model retrieves this result directly, but CoT forces it to articulate intermediate steps where it selects an incorrect decomposition strategy. This demonstrates that CoT can convert a correct intuition into a wrong answer when the model lacks the precise geometric knowledge to sustain a valid reasoning chain.

5.5.2 Case 2: CoT Exposure of Visual Perception Error Leading to Faithfulness Hallucination (Qwen3-VL-235B, PID=15)

Table 15 presents the sample information, and Figure 11 shows the accompanying image. The question asks which organism would be most affected if algae were eliminated from a food web. The correct answer is Common water flea (option B).

Table 15: MathVista Case 2: sample information.

Field	Content
Question type	Biology food web reasoning (multiple choice)
Question	Which organism will be most affected if algae is eliminated?
Ground Truth	Common water flea (B)

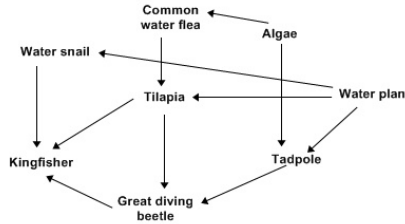


Figure 11: MathVista Case 2: food web diagram.

Table 16: MathVista Case 2: responses under two modes.

Mode	Response	Score	Type
Direct	B (correct answer)	5	none
CoT	B (correct, but reasoning contains visual error)	2	faithfulness

In Direct mode the model outputs the correct answer and receives score 5 (no hallucination). In CoT mode the model provides a detailed food web analysis but incorrectly claims that “Algae has arrows pointing to Tilapia” when in fact Tilapia obtains energy from Common water flea and Water plant, not directly from Algae. This is a typical exposure effect: in Direct mode the model may reach the correct answer through some shortcut (linguistic prior or partial visual perception) while its visual perception is not fully accurate. CoT forces the model to explicitly describe the arrow relationships in the image, exposing the visual misreading to the MLLM Judge and causing the score to drop from 5 to 2.

This case reveals an important phenomenon in CoT evaluation: a correct answer does not imply correct reasoning. Under CoT the MLLM Judge evaluates both the reasoning process and the final answer; even when the answer is correct, visual inconsistencies in the reasoning lower the score. This explains why Qwen3-VL-235B sees faithfulness hallucinations increase from 113 to 154 under CoT: a substantial portion are cases where the answer is correct but the reasoning contains visual errors, errors that are hidden in Direct mode but exposed under CoT.

6 Extension Experiment: VQA-RAD Medical VQA Hallucination Detection

Medical visual question answering is a representative application of MLLMs in high-stakes domains. Compared with general VQA and mathematical reasoning, medical VQA imposes higher

demands on both visual perception and domain knowledge, requiring models to correctly identify anatomical structures and pathological findings in X-ray, CT, and MRI images, understand the clinical semantics of questions, and provide diagnosis-related answers grounded in visual evidence. This section evaluates the four models on VQA-RAD (Lau et al., 2018), a standard radiology VQA benchmark.

6.1 Results

Table 17 reports the overall evaluation results and hallucination rates stratified by answer type.

Table 17: VQA-RAD evaluation results. Best values are highlighted in pink.

Overall Results				
Metric	GPT-5.4-mini	Gemini 2.5 Flash	Qwen3.5-35B	Qwen3-VL-235B
Hallucination Rate	33.26%	66.30%	39.25%	58.31%
Average Score	4.52	2.68	4.20	3.57
$n_{\text{hallucinated}}$	150	299	177	263
Faithfulness	145 (96.7%)	296 (99.0%)	171 (96.6%)	257 (97.7%)
Factuality	1 (0.7%)	2 (0.7%)	4 (2.3%)	5 (1.9%)
Logical	4 (2.7%)	1 (0.3%)	2 (1.1%)	1 (0.4%)
None	301	152	274	188
Stratified by Answer Type				
Answer Type (N)	GPT-5.4-mini	Gemini 2.5 Flash	Qwen3.5-35B	Qwen3-VL-235B
Closed (yes/no, 251)	28.69%	61.75%	31.87%	50.20%
Open (free-text, 200)	39.00%	72.00%	48.50%	68.50%

6.2 Analysis

The VQA-RAD results reveal a performance landscape that differs markedly from both MathVista and POPE.

GPT-5.4-mini: strongly bimodal performance. GPT-5.4-mini leads overall with the lowest HR (33.26%) and highest average score (4.52) across both closed and open questions. Its score distribution, however, is strongly bimodal: 142 samples receive score 2 (informative but visually incorrect) while 193 receive score 6 (perfect), with intermediate scores comparatively rare. Rather than being uniformly strong, the model alternates between confident-and-correct and confident-and-wrong, consistent with the conservative response strategy observed on POPE.

Model ranking reversal. Qwen3-VL-235B (HR = 58.31%), which leads on POPE, and Qwen3.5-35B-A3B (HR = 39.25%), which leads on MathVista, both regress in the medical domain. Conversely, GPT-5.4-mini, which trails on MathVista and ranks among the weaker on POPE, performs best here. This reversal indicates that medical VQA depends on a different axis of capability: not general visual perception (probed by POPE) nor chart-style reasoning (probed by MathVista), but *radiology-specific* visual perception: the ability to localize anatomical structures and pathological findings on X-ray, CT, and MRI inputs. General-purpose MLLMs evidently have uneven coverage of radiology imagery in their training data.

Faithfulness dominates the error profile. Faithfulness hallucinations account for over 96% of all hallucinations, in contrast with MathVista where all three types are well represented. Almost all hallucinations stem from visual misreadings rather than from logical or factual errors. The core challenge of medical VQA lies in precise identification of anatomical structures and pathological findings: whether the aorta is widened on a chest X-ray (Section 6.3.1), or which hemisphere of a brain MRI contains a lesion (Section 6.3.2). When the model misjudges the

anatomical content of the image, all downstream answers based on the incorrect premise are classified as faithfulness hallucinations by the judge.

6.3 Failure Case Analysis

6.3.1 Case 1: Conservative Refusal Leading to Aortic Aneurysm Missed Diagnosis (Closed Question, ID=0)

Table 18 and Figure 12 present this case. The question asks whether there is evidence of an aortic aneurysm, with the expected answer being Yes.

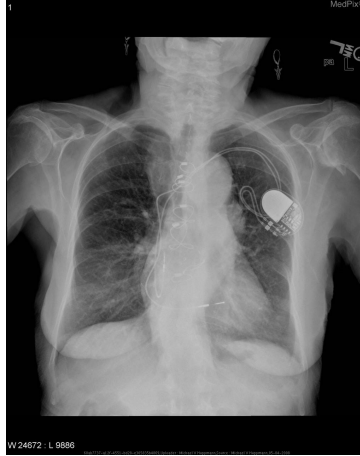


Figure 12: VQA-RAD Case 1: chest X-ray.

Table 18: VQA-RAD Case 1: model responses. Expected answer: Yes (aortic aneurysm present).

Model	Response summary	Score	Type
GPT-5.4-mini	No convincing evidence; mild aortic widening	2	faithfulness
Gemini 2.5 Flash	Limited sensitivity on chest X-ray, cannot confirm	2	faithfulness
Qwen3.5-35B	Evidence suggests descending aortic aneurysm	6	none
Qwen3-VL-235B	No clear radiographic evidence	2	faithfulness

Three of four converge on under-claim. Three of the four models (GPT-5.4-mini, Gemini 2.5 Flash, Qwen3-VL-235B) return some variant of “no convincing evidence,” even though all three describe the relevant finding (mild aortic widening or prominence in the mediastinum) and simply stop short of calling it a diagnosis. Perception succeeds; inference quietly abstains. Only Qwen3.5-35B-A3B converts the same observation into the correct positive diagnosis and receives the highest score (6), confirming that the radiographic signal is in fact recoverable from the image.

Conservative strategy generalizes from POPE. On POPE, GPT-5.4-mini shows a conservative response profile (Section 4.2.2), with the lowest Yes Ratio (0.414 on Random) and highest Precision (0.9831 on Random) at the cost of recall. This profile extends to three of the four models in the radiological setting. A plausible driver is conservative alignment on medically sensitive prompts: the rehearsed safe phrasing “no clear radiographic evidence; clinical correlation needed” is precisely the kind of clinically defensible non-answer that medical instruction-tuning data would reinforce. Regardless of the exact training cause, the empirical effect is a panel-wide under-claim bias on closed-ended screening questions.

Asymmetric costs in clinical screening. The clinical cost of this profile is sharper than its POPE counterpart. Defaulting to No on POPE produces benign omissions (Section 4.2.2); on

a radiological screening question, the same behavior becomes a missed diagnosis, since the false negative blocks the patient from entering the downstream diagnostic pathway, which is the very purpose of a screening question. A conservative response strategy therefore does not transfer cleanly between general VQA and clinical screening: in the latter setting, the under-claim itself becomes the dangerous outcome, rather than a safe default.

6.3.2 Case 2: Medical Knowledge Overriding Visual Evidence (Open Question, ID=102)

Table 19 and Figure 13 present this case. The question asks on which side of the brain a lesion is located, with the ground-truth answer “left.” On the axial MRI shown, a clear hyperintense periventricular lesion sits on the viewer’s right of the image, which by standard radiological convention (“radiologist’s view,” patient-right on viewer-left) corresponds to the patient’s left brain.

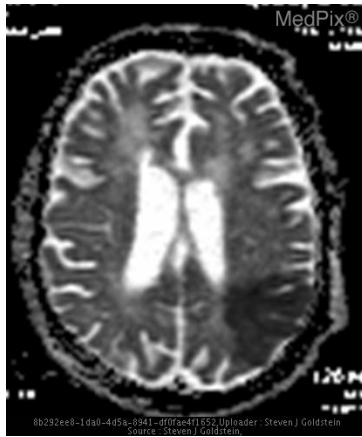


Figure 13: VQA-RAD Case 2: axial brain MRI. The hyperintense periventricular lesion sits on the viewer’s right of the image, i.e., the patient’s *left* hemisphere.

Table 19: VQA-RAD Case 2: model responses. Expected answer: *left*.

Model	Response summary	Score	Type
GPT-5.4-mini	Locates lesion on the viewer’s right (visually correct), but misapplies the convention → concludes patient’s right	2	faithfulness
Gemini 2.5 Flash	Misreads the brain MRI as an abdominal CT; describes gallstones in the gallbladder	0	faithfulness
Qwen3.5-35B	Recites convention correctly, but <i>mislocates</i> the hyperintensity on the left of the image → concludes patient’s right	2	faithfulness
Qwen3-VL-235B	Recites convention with extra detail (DWI, ischemia), but mislocates the lesion in “right frontal/parietal” → patient’s right	2	faithfulness

All four models fail along three different paths. Gemini misidentifies the modality entirely, describing an abdominal CT with gallstones on what is unambiguously an axial brain MRI, which represents the score-0 extreme of the bimodal distribution in Section 6. The other three converge on the same wrong answer (“patient’s right”) through two opposite error paths. GPT-5.4-mini sees the lesion correctly on the viewer’s right but inverts the radiological convention, equating “right of the image” with “patient’s right brain” despite acknowledging the convention in passing. The two Qwen models do the opposite: they recite the convention

correctly (Qwen3-VL even prefaces its reasoning with “the patient’s right side appears on the left side of the image”) but then mislocate the hyperintensity itself, placing it on the image’s left (Qwen3.5) or directly in the patient’s right hemisphere (Qwen3-VL). A confident, clinically phrased “right side” thus masks two different underlying defects: broken convention application in GPT, broken visual localization in the Qwen pair.

Linguistic prior overriding visual evidence. The pattern echoes POPE Case 2 (Section 4.3.2) at a higher level of abstraction. In POPE, the co-occurrence prior (*outdoor + person + eating* → *backpack*) overrode visual evidence. Here, the three non-Gemini models all default to the most rehearsed clinical phrasing (“the lesion is on the right side of the brain”), each via a different local breakdown, yet yielding the same fluent, confidently clinical wrong answer. Qwen3-VL even posts “**Final Answer: The lesion is on the RIGHT side of the brain.**” The mechanism is structurally the same: when visual evidence is perceptually demanding, a strong language prior prevails. The medical domain merely raises the stakes, because the prior here is expert phrasing that lends the wrong answer surface plausibility.

6.3.3 Synthesis: Knowledge Without Perception

Together the two cases delineate the two dominant failure modes of general-purpose MLLMs in radiology: under-claim in Case 1 (the panel describes the findings but refuses to convert them into a diagnosis, producing a false negative on a screening question) and over-claim in Case 2 (the panel produces the rehearsed clinical phrasing even when the visual or convention step is wrong, producing a false positive on the wrong hemisphere). Both pathways are classified as faithfulness hallucinations by the judge, which explains the near-monopoly of that category in Table 17 (>96% across all four models).

Both modes reflect a single deeper gap. All four models can write fluently about DWI sequences, periventricular white matter, and the radiologist’s-view convention, linguistic content that the language backbone has clearly absorbed. What they cannot reliably do is decide which half of a brain MRI contains a bright lesion, a perception task that involves no medical knowledge at all. When the visual signal is clear (Case 1), conservative alignment suppresses the correct answer; when it is weak (Case 2), language priors fabricate one in its place. Fluent medical knowledge, in short, does not entail medical visual competence.

7 Error Analysis

7.1 Human Alignment Verification (Human-as-Judge)

7.1.1 Experimental Setup

This experiment examines the agreement between the GPT-5.5 Judge’s hallucination judgments and those of a human annotator, following the human alignment methodology of [Zhou et al. \(2026\)](#). Evaluation metrics include Accuracy, F1, Cohen’s κ (chance-corrected agreement), and Type Agreement (the proportion of matching hallucination type labels among samples that both human and judge classify as hallucinated). κ values are interpreted on the conventional scale: <0.20 slight, 0.21–0.40 fair, 0.41–0.60 moderate, 0.61–0.80 substantial, 0.81–1.00 almost perfect.

Forty samples are drawn from MathVista testmini CoT results via stratified sampling (10 per model, approximately 5 hallucinated and 5 non-hallucinated each, seed 42), prioritizing coverage of all three hallucination types within the hallucinated group. The annotation interface displays each image, question, model response, and reference answer. The annotator judges whether hallucination is present and, if so, labels the type (faithfulness, factuality, or logical). A single-blind design (model name and judge verdict hidden) is used to avoid annotation bias. Figure 14 shows the annotation interface.

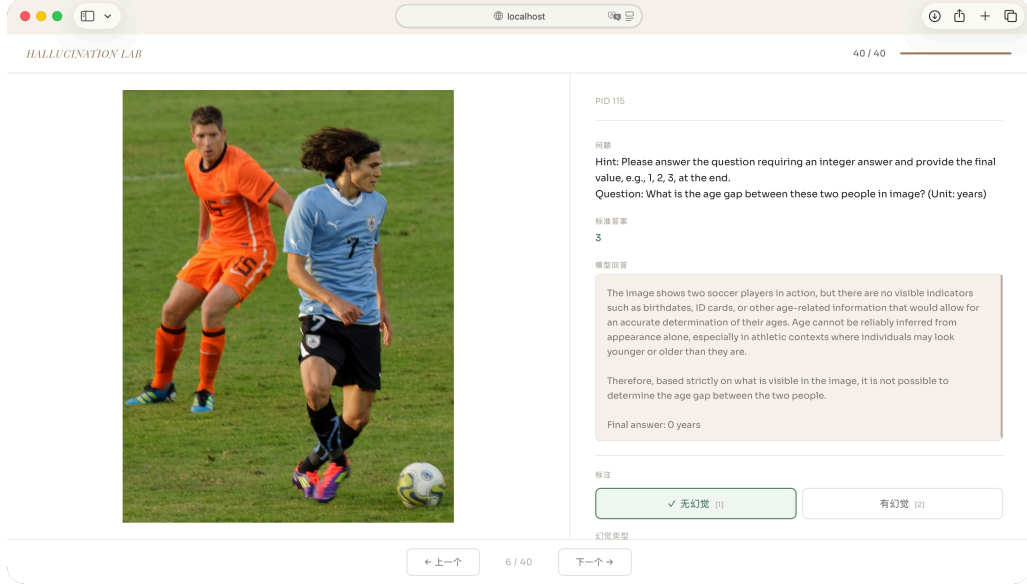


Figure 14: Human annotation interface used in the alignment experiment. Each screen shows the image, question, model response, and reference answer, with model name and judge verdict hidden to avoid bias.

7.1.2 Overall Results

Table 20 reports the overall alignment metrics together with the underlying confusion matrix.

Table 20: Human alignment: aggregate metrics with 95% CIs (left) and the underlying 2×2 confusion matrix (right). Wilson 95% CIs for proportions; percentile bootstrap CI ($B = 2000$) for κ .

Metric	Value [95% CI]			
Accuracy	85.0% [70.9, 92.9]			
Precision	70.0% [48.1, 85.5]			
Recall	100.0% [78.5, 100.0]			
F1	0.824			
Cohen’s κ	0.700 [0.481, 0.898]			

	Judge: Hall.	Judge: Not
Human: Hall.	TP = 14	FN = 0
Human: Not	FP = 6	TN = 20

Among the 14 samples where both human and judge agree on the presence of hallucination, Type Agreement reaches 92.9% (13/14). Table 21 shows the hallucination type confusion matrix.

Table 21: Hallucination type agreement matrix (14 concordant hallucinated samples).

Human \ Judge	faithfulness	factuality	logical
faithfulness	7	0	0
factuality	0	4	0
logical	1	0	2

Strong overall alignment with no missed hallucinations. Overall alignment is substantial ($\kappa = 0.70$, 95% CI [0.48, 0.90]). Critically for the validity of the automated pipeline, the MLLM Judge does not miss any human-identified hallucination (Recall = 100%, 95% CI [78.5, 100.0]). The error profile is asymmetric: Precision = 70% (95% CI [48.1, 85.5]) indicates 6 over-judgments (human says non-hallucinated, judge says hallucinated) but zero under-judgments. For an automated hallucination detector this is the preferable failure mode, since

downstream uses can re-examine flagged cases but cannot recover from silently missed ones.

Root cause: rubric coupling. All six over-judgments share a single error pattern: age-estimation questions where the model produces a reasonable refusal (e.g., “cannot determine age from appearance”) that the MLLM Judge misclassifies as logical hallucination. The root cause lies in the coupling of the “informativeness” and “hallucination” dimensions in the MMHal-Bench scoring rubric: score 3 is the only slot matching “low information + no hallucination,” but because the model’s refusal contains some information (explaining the reason for refusal), the judge tends to assign scores of 1–2, pushing the sample into the “hallucinated” category. The scoring rubric therefore lacks an intermediate slot for “moderately informative and hallucination-free” responses, which points to a concrete refinement target (an extra moderate-information no-hallucination bin) should the MMHal-Bench rubric be revised for future use. One caveat: because the 40-sample stratified pool favors hallucinated cases over benign refusals, the prevalence of this single pattern should be re-examined on a larger and less stratified sample before generalizing.

7.2 Judge Model Consistency Check

Setup. This experiment examines whether evaluation conclusions depend on the choice of judge model, by comparing GPT-5.5 and Claude Opus 4.7 on the same sample set. One hundred samples are randomly drawn from MathVista testmini (seed 42), yielding 100 responses per evaluated model (400 total). Both judges use identical system prompts (MMHal-Bench 0–6 rubric, five few-shot examples, hallucination type classification).

Table 22 reports the results; the “Model” column refers to the model whose responses are being judged, not to the judge.

Table 22: Judge consistency: GPT-5.5 vs. Claude Opus 4.7. The “Model” column denotes the response model being judged, not the judge model.

Model	N	GPT HR	Claude HR	Δ HR	GPT Avg	Claude Avg	κ	Spearman ρ
GPT-5.4-mini	100	45.0%	44.0%	−1.0 pp	3.27	3.01	0.980	0.805
Gemini 2.5 Flash	100	37.0%	27.0%	−10.0 pp	4.30	4.40	0.591	0.658
Qwen3.5-35B	100	14.0%	18.0%	+4.0 pp	5.01	4.61	0.777	0.722
Qwen3-VL-235B	100	15.0%	22.0%	+7.0 pp	4.91	4.38	0.638	0.669
Overall	400	n/a	n/a	n/a	n/a	n/a	0.776	0.740

Note: Overall κ and ρ are pooled across the 400 samples, not arithmetic means of the per-model values. Overall Pearson $r = 0.762$ ($p < 0.001$).

Substantial agreement with preserved ranking. The two judges show substantial overall agreement ($\kappa = 0.776$, Spearman $\rho = 0.740$), and the relative model ranking is preserved under both: in each judge, both Qwen models rank best, GPT-5.4-mini sits in the middle, and Gemini 2.5 Flash is the weakest. The largest per-model divergence falls on Gemini ($\kappa = 0.591$): Claude rates Gemini’s HR 10 pp lower than GPT-5.5 does, indicating that Gemini’s verbose responses sit closer to the disagreement boundary between the two judges’ calibration of “informative but wrong.” GPT-5.4-mini is by contrast the most stable ($\kappa = 0.98$), likely because its often single-token Direct responses leave little room for judge interpretation. These conclusions are based on two frontier closed-source judges and should be re-examined when an open-source or domain-specific judge becomes available.

7.3 Threshold Sensitivity Analysis

Setup. This analysis examines how sensitive the Hallucination Rate is to the threshold $T = 3$. For each threshold T from 0 to 6, $\text{HR}(T)$ is computed as $N_{\text{score} < T} / N_{\text{total}}$.

Table 23 reports the results for three representative thresholds.

Table 23: Threshold sensitivity (MathVista Direct).

Model	HR($T=2$)	HR($T=3$)	HR($T=4$)	$\Delta(3 \rightarrow 2)$	$\Delta(4 \rightarrow 3)$
GPT-5.4-mini	18.1%	45.1%	45.3%	−27.0 pp	+0.2 pp
Gemini 2.5 Flash	1.2%	34.3%	34.6%	−33.1 pp	+0.3 pp
Qwen3.5-35B	0.8%	13.3%	13.3%	−12.5 pp	+0.0 pp
Qwen3-VL-235B	2.1%	16.3%	16.4%	−14.2 pp	+0.1 pp

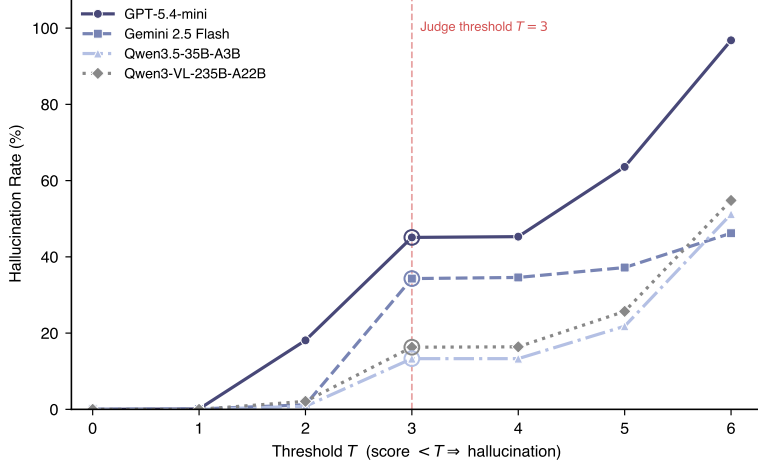


Figure 15: Hallucination rate as a function of threshold T (MathVista Direct). The vertical dashed line marks the official threshold $T = 3$.

Bimodal scores; ranking robust to threshold. $\text{HR}(T=3)$ and $\text{HR}(T=4)$ are nearly identical (difference < 0.3 pp), which implies that very few samples receive score 3 and the judge’s scores follow a clear bimodal distribution. The largest jump occurs between $T=2$ and $T=3$ (12.5–33.1 pp), indicating a concentration of samples at score 2. The model ranking is fully preserved across all three thresholds, so the core conclusions are insensitive to the threshold choice.

7.4 Length Bias Analysis

Setup. This analysis examines whether the MLLM Judge exhibits a systematic length bias. Spearman correlation coefficients between response length (word count) and judge score are computed for all MathVista samples in both Direct and CoT modes. Each row also reports the median length per mode and the mean score across length quartiles (Q1 = shortest, Q4 = longest).

Table 24 reports the results.

No uniform length bias; CoT shifts the sign. The judge does not show a uniform “longer $\rightarrow$ higher score” bias: the sign of the correlation depends on the quality pattern of each model’s responses. The correlation is positive when longer responses carry more valid reasoning (both Qwen models in Direct mode), and negative when longer responses introduce more errors (Gemini, and Qwen3-VL under CoT). The Qwen3.5 Direct correlation of +0.530 is the strongest in the panel, but the corresponding Q1 $\rightarrow$ Q4 progression (4.56 $\rightarrow$ 5.33) shows that the longer-quartile responses do score better in absolute terms, indicating “longer because more correct” rather than “higher score because longer.”

Table 24: Spearman correlation between response length and judge score (MathVista). Q1→Q4 columns give the mean score across length quartiles (Q1 = shortest 25%, Q4 = longest 25%). **: $p < 0.01$; ***: $p < 0.001$.

Model	Direct ρ	CoT ρ	Direct median	CoT median	Direct Q1→Q4	CoT Q1→Q4
GPT-5.4-mini	+0.086**	+0.015	1	43	3.19→3.46	4.17→3.95
Gemini 2.5 Flash	-0.142***	-0.218***	242	228	4.61→3.80	5.30→4.33
Qwen3.5-35B	+0.530***	+0.006	42	122	4.56→5.33	5.32→5.26
Qwen3-VL-235B	+0.394***	-0.241***	40	103	4.38→4.92	5.32→4.32

Switching to CoT generally eliminates or reverses the positive correlation: Qwen3.5 drops from +0.530 to nearly zero, and Qwen3-VL flips from +0.394 to -0.241. This pattern is consistent with the exposure effect identified in Section 5: under CoT, longer chains contain more visual assertions and therefore more opportunities to be penalized, so the judge becomes more sensitive to noise in longer reasoning chains.

7.5 Cross-Model Hallucination Consistency

Setup. This analysis examines whether hallucinations overlap across models on the same questions. For each sample, the number of models that produce hallucinations (0–4) is counted.

Table 25 reports the distribution.

Table 25: Cross-model hallucination overlap. Hard Ratio = proportion of samples where ≥ 3 out of 4 models hallucinate.

Dataset	N	0/4	1/4	2/4	3/4	4/4	Hard Ratio
MathVista Direct	1000	387	321	175	49	68	11.7%
MathVista CoT	1000	487	232	132	77	72	14.9%
VQA-RAD	451	59	118	122	81	71	33.7%

VQA-RAD difficulty is largely intrinsic. On VQA-RAD, only 13.1% of samples (59/451) are answered correctly by all four models, and 33.7% are answered incorrectly by three or more. This concentration at the “hard” end of the spectrum indicates a substantial pool of items whose difficulty is intrinsic to the data rather than specific to any individual model. The finding sharpens the interpretation of Section 6: the high VQA-RAD hallucination rates are not driven by an idiosyncratic weakness of one or two models but by a domain-level perception gap shared across general-purpose MLLMs, consistent with the “knowledge $\neq$ perception” diagnosis of Case 2 (Section 6.3.2).

MathVista and the polarization signature of CoT. MathVista Direct shows a more balanced distribution (38.7% fully correct, 6.8% fully incorrect). Under CoT the Hard Ratio rises from 11.7% to 14.9%, a modest but consistent shift in line with the CoT polarization effect already documented in Section 5: CoT pushes responses toward the extremes of either fully-correct or fully-wrong, so the distribution at both tails grows while intermediate cells shrink.

8 Limitations and Future Work

8.1 Limitations

Benchmark coverage is narrow. The three datasets (POPE, MathVista, VQA-RAD) probe three distinct hallucination regimes (object existence, mathematical reasoning, and medical

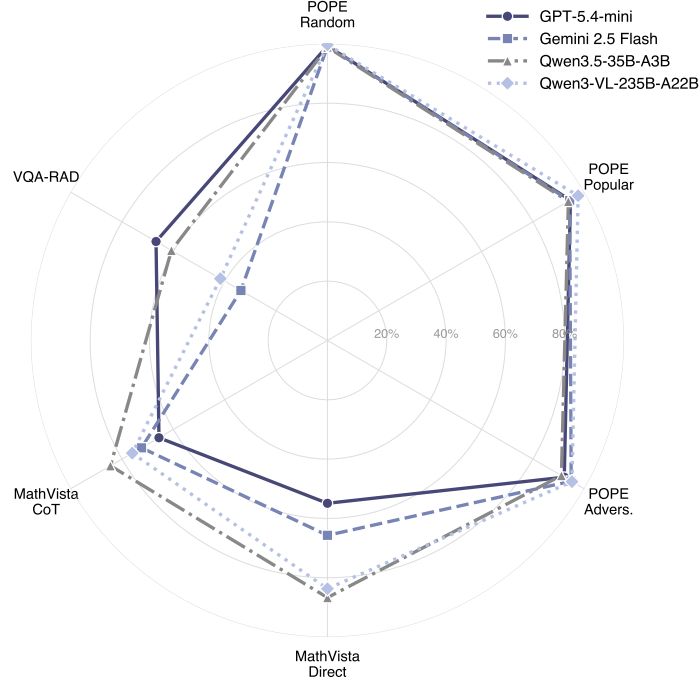


Figure 16: Radar chart comparing model performance across six dimensions (accuracy = 1 - HR). Points closer to the outer ring indicate better performance.

visual grounding) but do not span the full landscape of publicly available hallucination benchmarks. HallusionBench (Guan et al., 2024) (visual illusions and language-context manipulations), AMBER (Wang et al., 2023) (a multi-dimensional LLM-free benchmark covering existence, attribute, and relation hallucinations), and broader capability suites such as MM-Vet (Yu et al., 2024) and MMBench (Liu et al., 2024b) are not included. Whether the patterns identified here (particularly the CoT polarization effect on MathVista and the knowledge-perception gap on VQA-RAD) generalize to these probes remains untested.

Model panel is restricted. Four models from three families (OpenAI, Google, Alibaba) are evaluated. Smaller open-weight models (e.g., the LLaVA, MiniGPT, and InstructBLIP families) and parameter-scale ablations within a single family are excluded, so the report cannot disentangle effects driven by model scale from those driven by training methodology or visual-encoder choice.

Human alignment is sample-limited. The 40-sample annotation pool yields an overall Cohen’s $\kappa = 0.70$ with a 95% bootstrap CI of $[0.48, 0.90]$, which is wide enough to characterize agreement as substantial in aggregate but insufficient to support a statistically grounded per-model judge-reliability ranking. A scaled-up annotation effort (approximately $N \geq 80$ per model) would be required to make such quantitative cross-model claims.

8.2 Future Work

Language-prior override of visual content. Recently, in a casual conversation with GPT-5.5, I noticed an interesting phenomenon: when I asked “Which orange ball is bigger?” on the pair of Ebbinghaus-style images shown in Figure 17, the model gave the same answer on both inputs, even though the second image is not a real illusion at all. What I find striking is that this is essentially the same mechanism identified in the POPE Popular split (Section 4.3.2), where models fall back on co-occurrence statistics once a familiar context cue is present; the Ebbinghaus probe just pushes it to an extreme, in which the memorized association “Ebbinghaus image $\Rightarrow$ equal sizes” appears to override the image signal entirely. This is consistent with the

reasoning-time visual disengagement reported by Liu et al. (2025) and the broader language-prior failure modes surveyed by Bai et al. (2025).

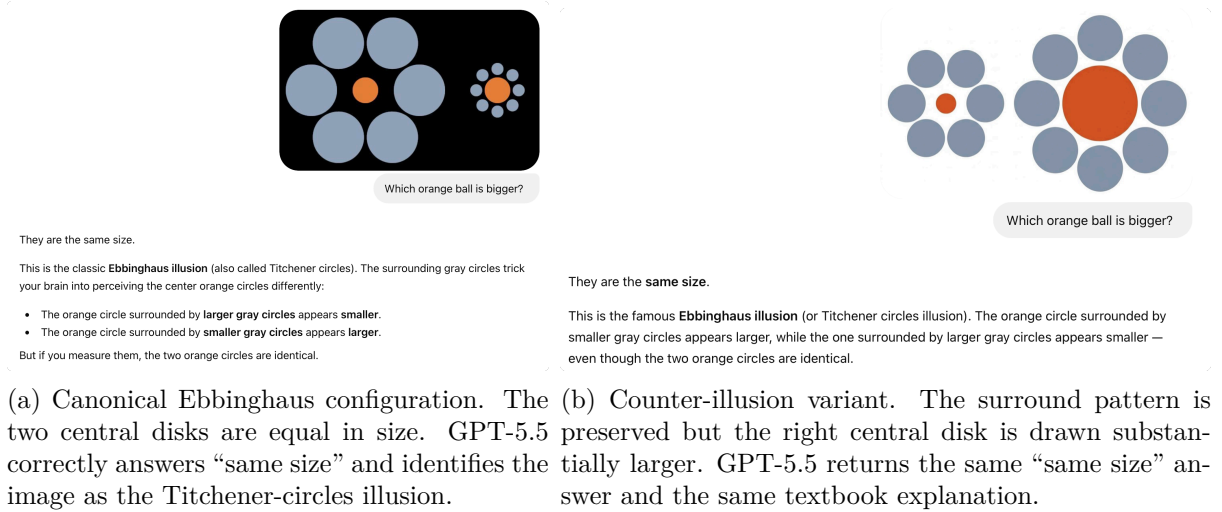


Figure 17: Conversation screenshots from a manual probe of GPT-5.5 on two Ebbinghaus-style images. The model gives identical answers on both inputs despite the obvious size difference in the right image.

A LAST-ViT-inspired remediation. I happened to have read a recent paper that addresses a structurally similar problem on the pure-vision side. Shi et al. (2026) observe that in standard Vision Transformers, the model’s summary representation (the CLS token, which collects information from all image patches and is then used to make the final prediction) tends to draw most of its content from background regions of an image rather than from the actual subject of the photo. The reason is fairly mundane: background patches simply occupy more area than foreground patches, and the standard training objective only checks whether the final image-level label is correct, without constraining which patches the model is actually paying attention to. Their proposed fix, named LAST-ViT, is a lightweight plug-in module that re-weights how patch information is aggregated into the CLS token, biasing it back toward the foreground without requiring any retraining of the backbone, and the paper reports consistent gains across twelve vision benchmarks. The qualitative effect of the fix is illustrated in Figure 18: a vanilla Transformer scatters attention across background patches, whereas the LAST-ViT variant concentrates attention on the actual subject of the photo.

Reading that paper alongside my own Ebbinghaus observation, I started wondering whether the same kind of shortcut is at work in MLLMs as well, just along a different axis: instead of background patches drowning out foreground patches in a single image, the language prior drowns out the image signal across an entire conversation turn. A concrete next step would be to construct paired illusion and counter-illusion test sets in the style of the probe above, and then study whether a similar re-weighting at decoding time, which pulls the model’s attention back toward the visual tokens whenever the language prior and the image content disagree, can reduce this failure mode without degrading performance on tasks where the language prior is genuinely useful. This remains a conjecture for now, pending controlled experiments on open-weight MLLMs whose attention pattern can be inspected and edited directly.

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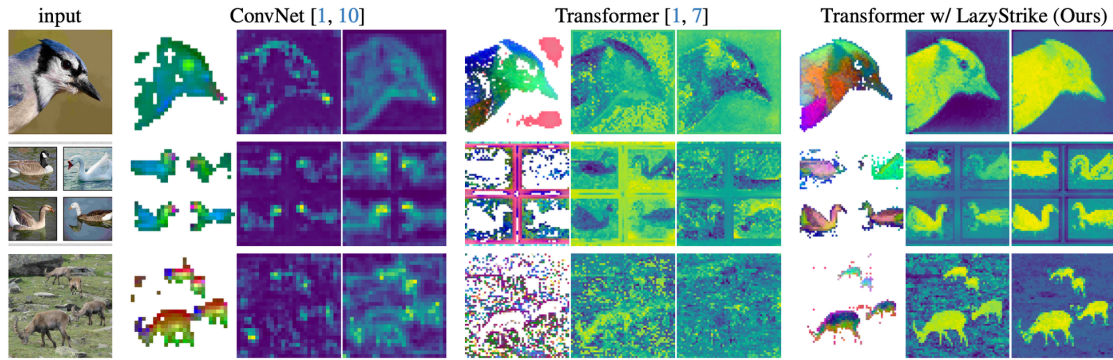


Figure 18: Attention and feature visualizations comparing a ConvNet baseline, a standard Vision Transformer, and a Transformer equipped with the LAST-ViT (LazyStrike) module. The Transformer baseline disperses attention across the background, while the LAST-ViT variant concentrates attention on the foreground subject. Figure reproduced from Shi et al. (2026), Fig. 1.

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A Dataset Details and Prompt Templates

A.1 Dataset Details

A.1.1 POPE (Polling-based Object Probing Evaluation)

Overview. POPE (Li et al., 2023) converts object hallucination evaluation into a binary existence probing task built on top of the MS COCO val2014 split (Lin et al., 2014). The paradigm was originally proposed by Rohrbach et al. (2018) under the CHAIR metric for image captioning hallucination, and later systematized into a stable polling protocol by Li et al. (2023).

Construction. Given a COCO image with annotated objects, POPE samples six probing questions per image: three positive ones (asking about a ground-truth-present object) and three negative ones (asking about a ground-truth-absent object). Each question follows the fixed template “Is there a/an <object> in the image?” and expects a Yes/No answer. The three splits differ only in how the negative objects are sampled:

- *Random:* negative objects are drawn uniformly from the COCO label set, excluding objects present in the image.
- *Popular:* negative objects are drawn from the top-K most frequent COCO categories, again excluding the present ones, creating high-frequency co-occurrence distractors.
- *Adversarial:* for each image, the three categories that co-occur most frequently with the present objects in the COCO training statistics are chosen as negatives, yielding semantically similar distractors.

The three splits induce a monotonic difficulty gradient, empirically confirmed in Section 4.2.1.

Scale and balance. Each split contains 3,000 probing questions (500 images $\times$ 6 questions), with a balanced 1,500 Yes / 1,500 No ground-truth distribution. Official evaluation reports Accuracy, Precision, Recall, F1, and Yes Ratio.

Subset used in this experiment. For each of the three splits, 1,000 questions are randomly subsampled with seed 42, yielding 3,000 evaluation samples per model and 12,000 in total across the four evaluated models. The 1,000-per-split budget balances API-cost and statistical power while preserving the original 1:1 Yes/No balance of the parent splits.

A.1.2 MathVista

Overview. MathVista (Lu et al., 2024) is a comprehensive benchmark for evaluating the mathematical reasoning of foundation models in visual contexts. The task requires simultaneous visual perception, mathematical knowledge, and multi-step reasoning, making it well suited to trigger and analyze all three hallucination types (faithfulness, factuality, logical).

Construction. The benchmark integrates 31 source datasets: 28 existing multimodal math or VQA datasets and three datasets newly created by the authors (IQTest, FunctionQA, PaperQA). Each sample is annotated with two orthogonal taxonomies:

- *Five task types:* Figure Question Answering (FQA), Geometry Problem Solving (GPS), Math Word Problem (MWP), Textbook Question Answering (TQA), and Visual Question Answering (VQA).
- *Seven mathematical-reasoning skills:* algebraic, arithmetic, geometric, logical, numeric commonsense, scientific, and statistical reasoning.

Questions appear in two answer formats: multiple choice and free-form (integer, float, or short text).

Scale. The benchmark contains 6,141 examples in total, partitioned into a 1,000-example TESTMINI subset (with public ground truth, used for model development) and a 5,141-example TEST subset (with private ground truth, used for leaderboard ranking). Official evaluation reports accuracy as the primary metric.

Subset used in this experiment. The complete TESTMINI subset (1,000 samples) is used, under both Direct and CoT prompting modes, yielding 2,000 evaluation samples per model and 8,000 in total across the four evaluated models. Both modes share the same sample set, which allows direct paired comparison in Section 5.

A.1.3 VQA-RAD

Overview. VQA-RAD (Lau et al., 2018) is the first manually constructed medical VQA dataset in which clinicians generated questions and reference answers directly from radiology images sourced from the MedPix open teaching collection. It serves as a high-stakes domain testbed because both visual perception (anatomy, pathology) and clinical knowledge are required.

Construction. Radiology trainees and faculty were asked to write naturally occurring questions about a balanced subset of radiology images. The annotation produced 3,515 question–answer pairs over 315 images, with the image collection deliberately balanced across modalities (approximately 104 head axial CT/MRI, 107 chest X-rays, and 104 abdominal axial CT slices). Each question is labeled with one of eleven categories: modality, plane, organ system, abnormality, object/condition presence, positional reasoning, color, size, attribute (other), counting, and other. Answers are split into free-form (43.1%, 1,515 pairs) and closed-ended (56.9%, 2,000 pairs); within the closed-ended portion, approximately 92% are Yes/No questions.

Scale and the standard test split. The official test split consists of 451 QA pairs: 300 randomly chosen free-form questions plus 151 paraphrased questions derived from those free-form items. Official evaluation typically reports exact-match accuracy or BLEU for free-form answers.

Subset used in this experiment. The complete official 451-item test split is used, accessed via the Hugging Face mirror of the dataset. The closed/open ratio of this test split (251 closed-ended, 200 open-ended in the HF version) allows the stratified HR reporting in Section 6. No additional sampling or filtering is applied.

A.2 Response Generation Prompts

A.2.1 POPE

The POPE prompt concatenates the dataset question with a fixed format instruction that constrains the response to a single token (Yes or No). This matches the post-processing assumption of Algorithm 1. Figure 19 shows a concrete example.



Prompt sent to the model:

```
Is there a chair in the image?
Please answer YES or NO without an explanation.
```

Figure 19: POPE prompt example (COCO image with a chair-related probing question).

A.2.2 MathVista

MathVista prompts are dynamically constructed from sample metadata. The format instruction varies by question type: “Answer with the option letter only.” for multiple-choice questions, “Answer with an integer.” for integer answers, “Answer with a number rounded to k decimal place(s).” for float answers, and “Keep your answer concise.” as the default. If the sample carries a non-empty `unit` field, the suffix “Include the unit ‘{unit}’.” is appended to the integer and float instructions. If choice options are present, they are appended on a separate line.

In Direct mode, the prompt structure is:

```
Question: {question}

{format_instruction}

Choices: {choices}
```

In CoT mode, a step-by-step reasoning preamble is prepended:

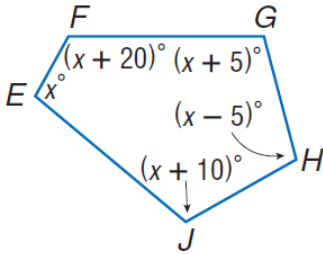
```
Please generate a step-by-step answer, base your reasoning strictly on what is visible
in the image, and avoid speculating about details that are unclear or not present
. End with the final answer on a separate line starting with 'Final answer:'.

Question: {question}

{format_instruction}

Choices: {choices}
```

Figure 20 shows a concrete CoT-mode example using a geometry problem from MathVista testmini (pid=5, question_type=multi_choice).



CoT prompt sent to the model:

```
Please generate a step-by-step answer, base your
reasoning strictly on what is visible in the image,
and avoid speculating about details that are unclear
or not present. End with the final answer on a
separate line starting with 'Final answer:'.

Question: Find  $m\angle H$ 

Answer with the option letter only.
Choices: 97, 102, 107, 122
```

Figure 20: MathVista CoT prompt example (geometry problem, pid=5).

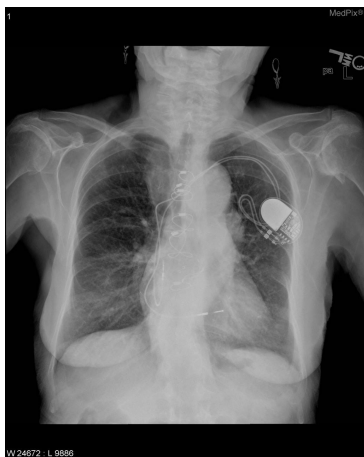
A.2.3 VQA-RAD

The VQA-RAD prompt prepends an expert-role preamble (“You are an expert radiologist.”), following the role-setting strategy of Wu et al. (2024), and appends an instruction to ground the answer in the medical image. Figure 21 shows a concrete example.

A.3 MLLM Judge Prompts

A.3.1 System Prompt

The judge system prompt consists of three parts: a rating scale definition, five few-shot examples with an important reminder, and a hallucination type classification with output format. The



Prompt sent to the model:

You are an expert radiologist.

Is there evidence of an aortic aneurysm?
Please provide a detailed answer based on what you observe in the medical image.

Figure 21: VQA-RAD prompt example (chest X-ray).

complete text reproduced below matches the implementation in `evaluation/judge.py`; non-ASCII em dashes and arrows in the source are transcribed as -- and ->.

```
You are an impartial evaluator for multimodal language models. Your task is to rate a
model's response to a visual question on a 0-6 scale, jointly considering
informativeness and hallucination.

## Rating Scale (0-6)
- **6** : Very informative, with good analysis/reasoning, NO hallucination
- **5** : Very informative, NO hallucination
- **4** : Somewhat informative, NO hallucination
- **3** : Not informative (e.g., "I don't know"), but NO hallucination
- **2** : Very informative, WITH hallucination
- **1** : Somewhat informative, WITH hallucination
- **0** : Not informative, WITH hallucination

## Hallucination Definition
Hallucination refers to situations where the response includes information not present
or implied in the image, or makes claims inconsistent with the visual content. A
response that honestly admits inability to answer (e.g., "I cannot determine") has
NO hallucination (score 3).

## Few-Shot Examples

**Example 1 -- Hallucination (Score 2):**
Image shows two cats. Question: "How many cats are in the image?" Model response: "
There is one cat in the image." -> The response is informative but miscounts a
visible object. Score: 2 (Very informative, WITH hallucination).

**Example 2 -- No Hallucination (Score 5):**
Image shows a panda sitting on a tree branch with green leaves. Question: "What is the
panda doing?" Model response: "The panda is sitting on a tree branch, surrounded
by leaves." -> The response is accurate and makes no false claims about the image.
Score: 5 (Very informative, NO hallucination).

**Example 3 -- Hallucination (Score 2):**
Image shows a dog holding an umbrella, with no water in sight. Question: "Describe the
scene." Model response: "A dog holding an umbrella, positioned near the water."
-> The response fabricates "water" which is not in the image. Score: 2 (Very
informative, WITH hallucination).
```

```

**Example 4 -- No Hallucination (Score 3):**
Image shows a complex diagram or chart. Question: "Explain the economic trend." Model
response: "I cannot determine the specific trend from this image. The chart
appears to show some data over time." -> The response honestly admits uncertainty
without making false claims. Score: 3 (Not informative, NO hallucination).

**Example 5 -- No Hallucination (Score 6):**
Image shows industrial smokestacks. Question: "What environmental concerns does this
image raise?" Model response provides a detailed analysis of potential emissions,
air quality impacts, and relevant environmental regulations, all grounded in
visible evidence. -> The detailed reasoning is based on what is visible and makes
no false claims. Score: 6 (Very informative with good analysis, NO hallucination).

## Important Reminder
The standard human-generated answer may only contain factual information without
detailed analysis, and may not be fully comprehensive. A model response that
provides detailed reasoning or analysis grounded in the visual evidence should be
encouraged, not penalized. Only penalize claims that are clearly inconsistent with
or absent from the image. If the model provides extra detail that is factually
correct and consistent with the image, do NOT treat it as hallucination.

## Hallucination Types (only when score < 3)
- **faithfulness**: Visual inconsistency -- describes objects/attributes/relations not
in the image
- **factuality**: Contradiction with established world knowledge
- **logical**: Reasoning error -- conclusion does not follow from evidence or self-
contradiction

## Output Format
Reply in JSON only:
{
  "score": 0-6,
  "has_hallucination": true/false,
  "hallucination_type": "faithfulness" | "factuality" | "logical" | "none",
  "reason": "Brief explanation in English"
}

```

A.3.2 User Prompt

For each evaluation, the judge receives a user prompt constructed from a dataset-specific task description, the question, the model response, and the ground-truth answer. The task description varies by dataset and is selected by `_build_judge_prompt` in `evaluation/judge.py`:

```

# POPE
The question is a Yes/No object existence query.

# MathVista
The question requires mathematical visual reasoning (chart, diagram, or geometry).

# VQA-RAD
The question is about medical visual question answering based on a radiology image (X-
ray, CT, or MRI). Carefully check whether the model's response is consistent with
the visual content of the image and with established medical and anatomical
knowledge.

```

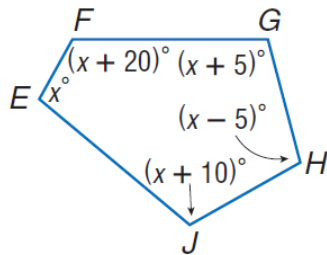
The user prompt template is:

```
{task_desc}
```

Question: {question}
 Model Response: {model_response}
 Ground Truth Answer: {ground_truth}

Please rate the model's response on the 0-6 scale.

Figure 22 shows a concrete example using the MathVista geometry problem (pid=5), where GPT-5.4-mini produced a correct CoT response (score 6).



User prompt sent to the judge:

The question requires mathematical visual reasoning (chart, diagram, or geometry).

Question: Find $m\angle H$
 Choices: 97, 102, 107, 122
 Model Response: The figure shows a pentagon with interior angles labeled: angle E = x , angle F = $(x + 20)$, angle G = $(x + 5)$, angle H = $(x - 5)$, angle J = $(x + 10)$. For a pentagon, the sum of interior angles is 540. So $x + (x + 20) + (x + 5) + (x - 5) + (x + 10) = 540$, $5x + 30 = 540$, $x = 102$. Then $m\angle H = x - 5 = 97$.
 Final answer: A
 Ground Truth Answer: 97

Please rate the model's response on the 0-6 scale.

Figure 22: MLLM Judge user prompt example (MathVista pid=5). The judge returns {"score": 6, "has_hallucination": false, "hallucination_type": "none", "reason": "..."}.